



Games of Symmetry

September 26, 2020



Turn Nim into a symmetry game

Let's return to our stones from last week.

1. 2 equal piles of 7. You may pick 1-4 stones in a row from the same pile. Winner takes the last stone.

Strategy: 2nd player mirrors the 1st player's moves with the opposite pile, and ends up picking up the last stone. Wherever there is a stone for the 1st player to pick up, there is one in the opposite pile for the 2nd player to pick up.

→ The 2nd player has the winning strategy.

Row of Stones, continued

2. One straight line of 15. You may pick 1-4 stones as long as they are in a row. Winner takes the last stone.

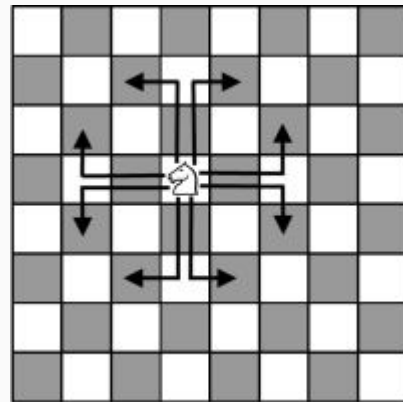
Strategy: 1st player takes the middle stone, then we are back to Case 1 with 2 equal piles of 7. 1st player uses symmetry to win against the 2nd player.

→ The 1st player has the winning strategy.

Symmetry on a chessboard - knights

Two players take turns placing knights on the squares of a chessboard, so that no knight attacks another. The player who is unable to place his/her next knight loses.

How does a knight move? Two squares in one direction and then one more square in a perpendicular direction.



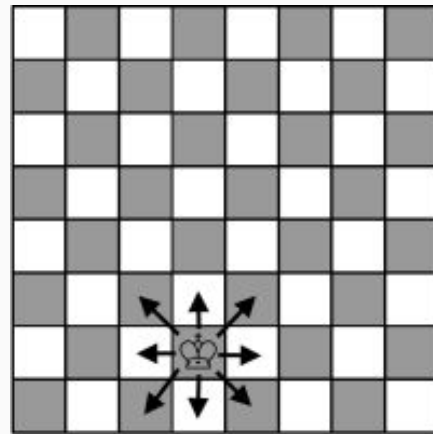
Symmetry on a chessboard cont. (knights)

Strategy: The 2nd player has a winning strategy, using symmetry around either a center line or point on the board. Any time the 1st player has a square to move to, the 2nd player will have the symmetrical square open.

Kings on a 9x9 chessboard

Two players take turns placing kings on the squares of a 9x9 chessboard, so that no king can attack another. The player who is unable to place his/her next king loses.

A king moves one square in any direction.



Kings on a 9x9 chessboard continued

Strategy: The 1st player places his/her king in the center square. Then the 1st player mirrors the 2nd player's moves using symmetry around the center square.

In other words, first make things even/equal, then play by symmetry. We used this in the row of 15 stones. This strategy is harder to see, but will come up a lot.

Breaking chocolate

Two players take turns breaking a piece of chocolate consisting of 5×10 squares. At each turn, they may break along the division lines of the squares. The player who first obtains a single square of chocolate wins.



Breaking chocolate continued

Strategy: The player who first breaks off a rectangle of width 1 loses. The first player will win if she/he plays optimally, by initially breaking the chocolate bar into two 5×5 pieces. After that, play symmetrically.

Idea: Again, whoever can turn this into a game where he mimics the other player symmetrically wins.